# HCI Phase 2: Solution Design and Prototype Development

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**Project:** RideBoard, a ride board web app linked from a WhatsApp bot, that helps IBA students find and offer carpools between home and campus.

This phase answers **RQ2 (design)** from Phase 1. It turns the Phase 1 findings into an evidence based design, following the chain the brief asks for:

**User problem → Research finding → Design requirement → HCI principle → Design decision → Prototype feature**

**Evidence codes:** Phase 1 survey (n = 26), poll (17 answers), interviews P1–P4, scenarios S1–S4. **Requirement codes:** F = functional (what the system does), I = interaction (how it behaves). **Scr** = wireframe screen in Section 7.

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## 1. Design requirements

Every requirement traces back to Phase 1 evidence, and every one maps to a prototype feature that can be observed in the Phase 3 usability test (for example, whether users notice the status line, or how many taps posting takes).

### 1.1 Functional requirements

| ID | User problem | Research finding | Design requirement | HCI principle | Design decision | Prototype feature |
|---|---|---|---|---|---|---|
| F1 | Requests get buried | 69% (18/26) say requests often go unanswered; "scrolls away" a top complaint | The system must keep ride requests and offers visible until they are filled or expire. | Visibility of system status | Requests and offers live on a ride board instead of scrolling away in the chat | Scr 4 ride board ("Ride needed" cards); Scr 1 home |
| F2 | Can't find who drives my route | Top voted problem in the Phase 1 poll (5 votes; P1 and P3 asked for filtering | The system must let users filter rides by origin, destination and time. | Recognition rather than recall | Filter chips for area, campus and time | Scr 4 filter chips |
| F3 | Can't tell what's still valid | P1 wants to know which offers are "still valid / have room" | The system must show the seats left on every ride and remove rides after departure. | Visibility of system status | Seat count on each card; auto-expire after departure time | Scr 4 "2 seats left" on cards |
| F4 | Last minute cancellations | P3 named cancellations as a failure | The system must notify the other party immediately when a ride is cancelled. | Feedback | Cancelling a confirmed ride sends a WhatsApp message to the other person, with a next step | Scr 6 cancel dialog; Scr 9 cancellation message |
| F5 | Recurring commutes are retyped every week | P1 and P3 prefer fixed schedules when timings are consistent | The system should let drivers publish a recurring weekly schedule. | Flexibility and efficiency of use | "Regular" option with day chips on the post form | Scr 5 Regular tab and day chips |
| F6 | No match in time, so the rider gives | P1 falls back to the group after ~10 min | When no driver is found, the system | Help users recognise, diagnose | After the timeout the status screen | Scr 2 no match state |

| ID | User problem | Research finding | Design requirement | HCI principle | Design decision | Prototype feature |
|---|---|---|---|---|---|---|
| | up and rides solo | with no reply; P4 books a Yango and posts if anyone wants to split the fare | must offer clear next steps instead of a dead end, including turning the rider's own booked trip into a shared ride offer. | and recover from errors | offers "Booking a ride anyway? Offer the empty seats", "Post to group" and "Keep waiting" | |
| F7 | Users prefer to stay in WhatsApp | Members voted to stay on WhatsApp rather than move to a new app (**23 votes)** | The system must minimise switching away from WhatsApp. | Match between system and the real world; consistency and standards | A WhatsApp bot handles sign up, pings, offers and updates; the web app, opened from a link, adds the board, posting and settings | Scr 7–9 WhatsApp chats |
| F8 | Ad hoc plans have no fixed time | 14 of 17 poll answers were same day or day before; riders often just want to get home (S1) | The system must let riders request either a time window or "any time". | Flexibility and efficiency of use | Time field with two options: "Between 7–8 pm" or "Whenever, just get me home" | Scr 3 "When?" toggle and time chips |
| F9 | Open time requests have no departure time | P2 waited hours and still got a ride, so a driver leaving later can still help | For open time requests, the system must let the driver propose a departure time, and confirm the ride only once | User control and freedom | Driver replies "leaving at 8:00"; rider gets Accept / Decline | Scr 8 time chips; Scr 1 offer card |

| ID | User problem | Research finding | Design requirement | HCI principle | Design decision | Prototype feature |
|---|---|---|---|---|---|---|
| | | | the rider accepts. | | | |
| F10 | Ad-hoc drivers never post, so nobody knows to ask them | P1 DMs "relevant" people she knows go her way; drivers in S1–S2 hadn't planned to post; S3 showed a driver passing through the rider's area was missed | The system must notify only drivers whose home area or usual route matches a request. | Aesthetic and minimalist design (relevant interruption s only) | Sign-up asks for home area plus "areas I pass through" (multi-select); pings go to matching drivers only | Scr 7 sign-up chat; Scr 10 "My areas" |
| F11 | Drivers' spare seats and schedules get buried in the group | Drivers post one off offers ("going to xyz at 5, if anyone wants to join", P1); schedule posts in the group are buried within days | The system must let drivers post one off ride offers (from the app or WhatsApp) and opt-in regular schedules, which riders can browse and request from. | Consistency and standards; recognition rather than recall | Board has "Today" (one off, expires after departure) and "Regular" tabs; "Request a seat" on every listing; WhatsApp quick command for one off posts | Scr 4 Today / Regular tabs; Scr 5 post form |
| F12 | Unclear whether a ride is a favour or paid | Scheduled offers in the group are usually paid; ad-hoc rides are usually favours | The system must let drivers mark any ride as free or paid, with free as the default. | User control; defaults as a gentle nudge | "Ask for a fare?" with "No, it's a favour" pre-selected and "Yes, Rs ___" as the alternative | Scr 5 fare control; Scr 8 fare question |

| ID | User problem | Research finding | Design requirement | HCI principle | Design decision | Prototype feature |
|---|---|---|---|---|---|---|
| F13 | Nobody knows whether a matched ride happened | P1 needs to know which offers are still valid; without an outcome, the board and status go stale | The system should confirm the outcome of each matched ride after departure. | Feedback (closing the loop) | One-tap WhatsApp follow-up: "Did your ride with Zainab happen? Yes / No" | Scr 9 follow-up message |

### 1.2 Interaction requirements

| ID | User problem | Research finding | Design requirement | HCI principle | Design decision | Prototype feature |
|---|---|---|---|---|---|---|
| I1 | Gives up and rides solo | P1 falls back to the group after ~10 min with no reply | The system must show the progress of a request at all times. | Visibility of system status; feedback | Status line on the request: "Pinged 4 · 1 declined · 3 pending" | Scr 1–2 status line |
| I2 | Posting takes effort | Members currently type free-text messages every time | Posting a request or a ride must take a few taps, with sensible defaults. | Flexibility and efficiency of use; recognition rather than recall | Campus pre-filled, area dropdown, one-tap time chips | Scr 3 new request; Scr 5 post a ride |
| I3 | Checking the same route over and over | P1 scans the last couple of hours of chat for her route each time | Filters must update results instantly and remember the user's usual route. | Flexibility and efficiency of use; user control | Live filtering; last used route saved as the default view | Scr 4 "Saved as your usual route" |
| I4 | Places named the way students say them | Requests in the group use area and campus names ("Gulshan", "main"), not addresses | The interface must use the place names students already use. | Match between system and the real world | Area list built from names used in the group; Main and City campus as fixed options | Scr 3, 5, 7 area lists |

| ID | User problem | Research finding | Design requirement | HCI principle | Design decision | Prototype feature |
|---|---|---|---|---|---|---|
| I5 | Plans change at the last minute | P3: last minute cancellations | Users must be able to edit or cancel a post easily, without starting over. | User control and freedom; error prevention | Edit and cancel on every post, with a confirmation step; cancelling triggers F4 | Scr 1 edit / cancel links; Scr 6 cancel dialog |
| I6 | Drivers are busy when a request arrives | S2: driver is getting into her car when pinged | Drivers must be able to answer a ping, including time and fare, in one or two taps without opening the app. | Flexibility and efficiency of use | WhatsApp quick replies: "Yes, leaving at…" (time chips) / "Can't"; "Pause pings for today" | Scr 8 quick replies; Scr 10 pause toggle |
| I7 | A published schedule reveals a routine | Members already post schedules openly in a women-only group; opening to all IBA widens who sees them | Regular listings must be opt-in, show only area and approximate time, and hide contact details until the driver accepts a request. | Privacy by design; user control | No street or live location; "Request a seat" instead of a phone number; contacts shared on acceptance | Scr 4–5 privacy notes; Scr 8 contact shared on acceptance |
| I8 | Comfort depends on who the other person is | P1: "same uni and gender helps"; P3: comfortable if she's from IBA | Drivers must be able to choose who sees their listings and receives their pings. | User control and freedom | Audience setting: women only / all IBA / people I've ridden with (gender self-declared at sign-up; see Section 11) | Scr 5 "Who can see this?"; Scr 10 |
| I9 | A fare that appears late feels like a trick | Fare expectations differ between ad-hoc and scheduled rides (F12) | Any fare must be visible before the rider requests or accepts a ride, never after. | Visibility; error prevention | Badge on every listing and offer: "Free · favour" or "Rs 300", shown again on the accept screen | Scr 1 and Scr 4 fare badges |

| ID | User problem | Research finding | Design requirement | HCI principle | Design decision | Prototype feature |
|---|---|---|---|---|---|---|
| I10 | Drivers must decide about a stranger in seconds | P1: "same uni and gender helps"; P3: comfortable if the rider is from IBA | Each ping and seat request must show one trust signal about the other person, with more detail on demand. | Progressive disclosure; recognition rather than recall | Ping shows "IBA verified · 2 rides together"; "See details" opens an optional rider profile | Scr 8 ping; Scr 1 and Scr 6 cards |

## 2. Personas

Both personas are composites of participants, with fictional names. The rider is well supported by data; the driver rests partly on assumptions, marked below.

### Rider: Mahnoor

- **Who:** Third-year student, lives in Gulshan-e-Iqbal, no car. Classes on both campuses.
- **How she commutes:** Mostly last-minute carpools from the group. Falls back to a Yango or inDrive when nothing works.
- **Current routine:** Scrolls the last couple of hours of the group for her route, DMs anyone relevant, waits about ten minutes, posts "Going to Gulshan at 5, anyone?", then books a ride if nobody replies. (P1, P2, P3)
- **Goals:** Get home on time without paying for a solo ride; know quickly whether a ride exists.
- **Frustrations:** Requests buried under other messages; no idea if an offer is still valid; people cancel last minute; few rides after 5 pm. (Poll, P1, P3)
- **Quote:** "A way to know all requests or offers for rides that are still valid or have room." (P1)

### Driver: Zainab

- **Who:** Final year student, lives in Gulistan-e-Jauhar, drives to campus most days with two or three empty seats. Her route passes through Gulshan.
- **How she offers rides:** Rarely plans ahead. Her leaving time depends on classes, events and meetings, so she replies to requests she happens to see in the group. (Assumption)
- **Goals:** Split fuel costs and help other students, without having to watch the group. (Assumption)
- **Frustrations:** Only sees requests if she happens to check the group at the right moment; doesn't want to be pinged about rides she can't give. (Assumption)

Zainab's assumptions will be checked with drivers in the Phase 3 evaluation.

## 3. Scenarios

S1–S3 are ad-hoc trips: neither person planned in advance, matching the poll (14 of 17 answers were same day or day-before). S4 covers the planned side: a regular schedule on the board.

### S1: Campus to home, same area

Mahnoor decides at the last minute to stay for a society event at Main campus. It ends at 7 pm, well after her usual ride left. At 6:45 she opens the app and posts a request: Main campus → Gulshan, between 7 and 8 pm. Zainab is at a different event that runs until 7:30, and she also lives near Gulshan. She hadn't planned to post anything, but she gets a WhatsApp ping: "Mahnoor needs Main → Gulshan between 7 and 8 pm. Going that way?" She taps **Yes, leaving at… 7:40** and keeps it as a favour. Mahnoor's status changes to "Zainab can take you at 7:40", she accepts, and they sort out the pickup on WhatsApp. After the ride, Mahnoor gets a one-tap message: "Did your ride with Zainab happen? Yes / No".

*Covers: F1, F7, F8, F9, F10, F13, I1, I6.*

### S2: Home to campus, same campus

Mahnoor's usual lift cancels at 7:30 am, and the cancellation arrives straight away on WhatsApp with "Want me to find another one?". She has a 9 am class at Main campus, so she taps **Yes, leaves by 8:15**. Zainab is just getting into her car near Gulshan, gets the ping, and taps **Yes, leaving at 8:00** without opening the app. Mahnoor sees the match within minutes instead of posting in the group and hoping someone is awake and checking it.

*Covers: F4, F8, F10, I1, I6.*

### S3: Campus to home, no driver available

Mahnoor needs to get from Main campus to Gulshan around 6:30 pm after a late group meeting. Nobody who lives in Gulshan is driving. Zainab lives in Gulistan-e-Jauhar but lists Gulshan as an area she passes through, so she is pinged too; she taps **Can't**, because she is leaving at 8:30. Mahnoor's status reads "Pinged 3 · 1 declined · 2 pending", and after ten minutes there is still no match. The status screen offers **Booking a ride anyway? Offer the empty seats**. She books a Yango, and her trip appears on the board as a shared ride to Gulshan with the split fare shown. A classmate going to Gulshan requests a seat.

*Covers: F6, F10, I1, I9. In version 1, matching used home area only, so Zainab was never pinged; see Section 10.*

### S4: A regular commute on the board

Areeba, a final year student with a fixed timetable, drives from Main campus to Gulistan-e-Jauhar every Monday and Wednesday around 5:30 pm, passing through Gulshan, with two empty seats. Instead of retyping it in the group every week, she posts a regular schedule in the app: Mon and Wed, about 5:30 pm, 2 seats, Rs 200 per seat, visible to women only. On Sunday night Mahnoor opens the board, filters Main → Gulshan, switches to **Regular**, and sees Areeba's listing with the Rs 200 badge, the area and the approximate time, but no phone number. She taps **Request a seat** for Wednesday. Areeba gets the request on WhatsApp and accepts, and only then are numbers shared.

*Covers: F2, F5, F11, F12, I7, I8, I9.*

## 4. Task models

Hierarchical task analysis (HTA; Annett, 2003) of the three core tasks. Each plan states the order of subtasks. The minimum tap counts become targets for the Phase 3 test.

### Task 1: Rider gets a ride (Mahnoor)

**0. Get a ride home**

1. Open RideBoard from the WhatsApp link
2. Create a request
    a. Confirm pick-up campus (pre-filled)
    b. Choose destination area
    c. Choose a time window or "Whenever"
    d. Tap "Find a ride"
3. Monitor the request status
4. Respond to an offer
    a. Check time, fare and trust signal
    b. Accept or decline
5. Coordinate pickup on WhatsApp
6. Confirm the ride happened (one tap)

**Plan 0:** do 1–3. If an offer arrives, do 4; if accepted, do 5 then 6. If there is no offer after ~10 minutes, choose a fallback: offer the empty seats, post to the group, or keep waiting (back to 3). **Plan 4:** 4.1 then 4.2; if declined, return to 3. **Minimum taps for step 2:** 3 (destination, time chip, Find a ride).

### Task 2: Driver responds to a ping (Zainab)

**0. Respond to a ride ping**

1. Read the ping (route, time, rider name, trust signal)
2. Decide whether you can help
    a. Optional: open rider details
3. Make an offer
    a. Choose leaving time
    b. Choose favour or fare
4. Wait for the rider's answer
5. Coordinate pickup on WhatsApp

**Plan 0:** do 1–2. If "Can't", stop. If "Yes", do 3.1 → 3.2 → 4; if the rider accepts, do 5. **Minimum taps:** 3 (Yes, time, favour), with no app needed.

### Task 3: Driver posts a regular schedule (Areeba)

**0. Post a regular ride**

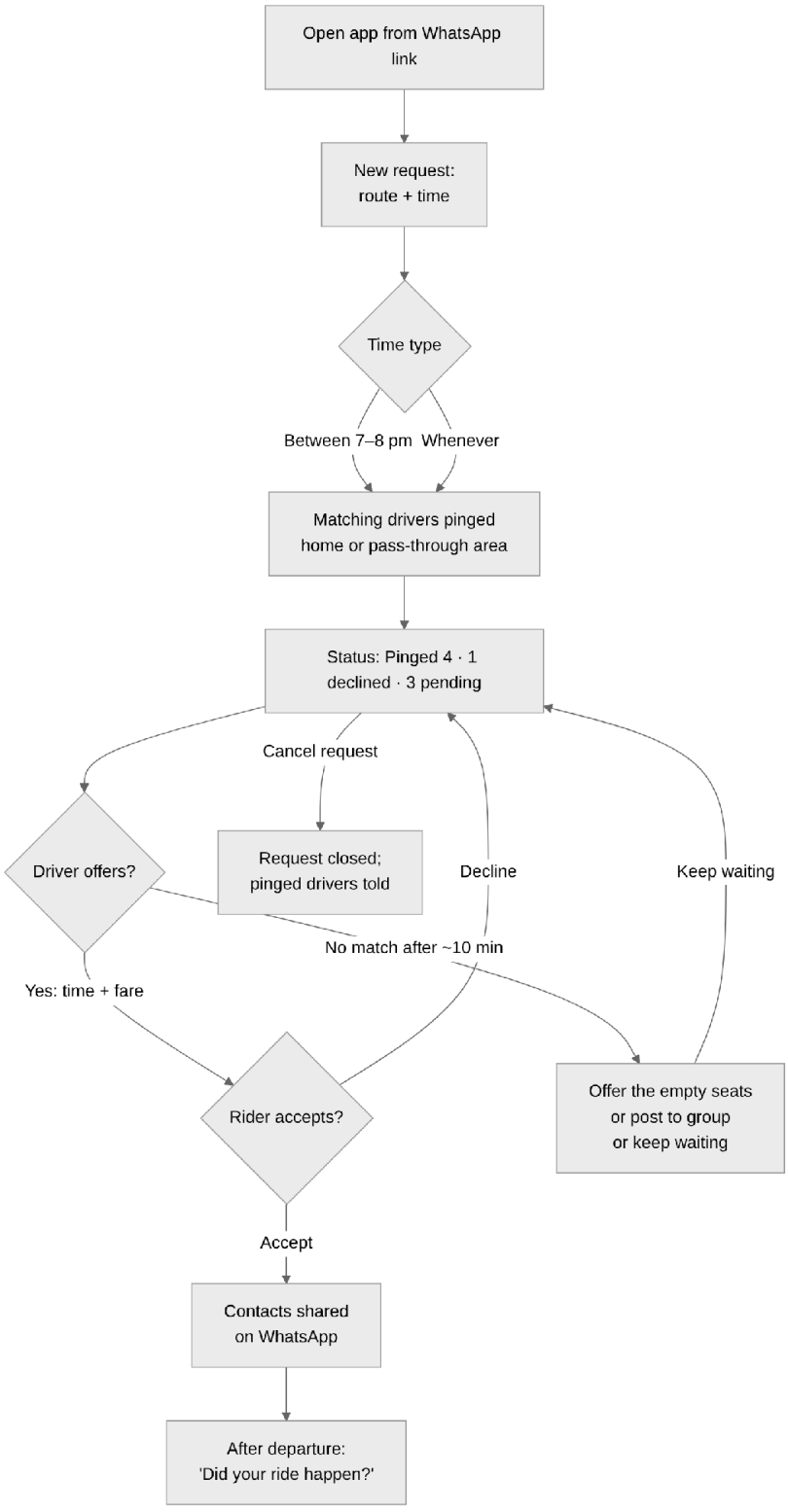
1. Open "Post a ride"
2. Choose "Regular"
3. Set days, route, pass through areas, approximate time and seats

4. Choose favour or fare
5. Choose audience
6. Post the ride
7. Handle seat requests (accept or decline on WhatsApp or in My rides)

**Plan 0:** do 1–6 in order; do 7 whenever a request arrives.

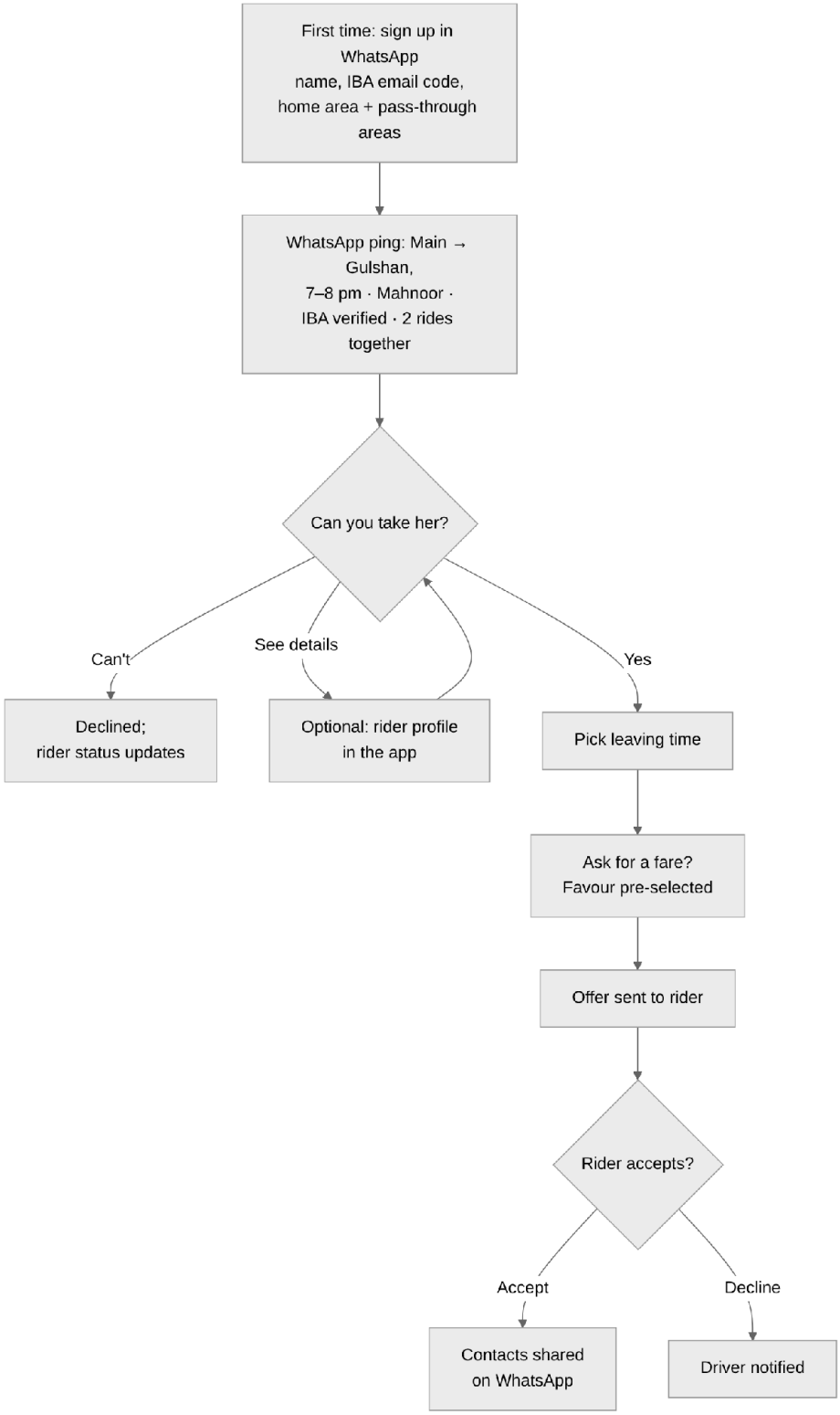
## 5. User flows

### Flow 1: Rider makes an ad-hoc request



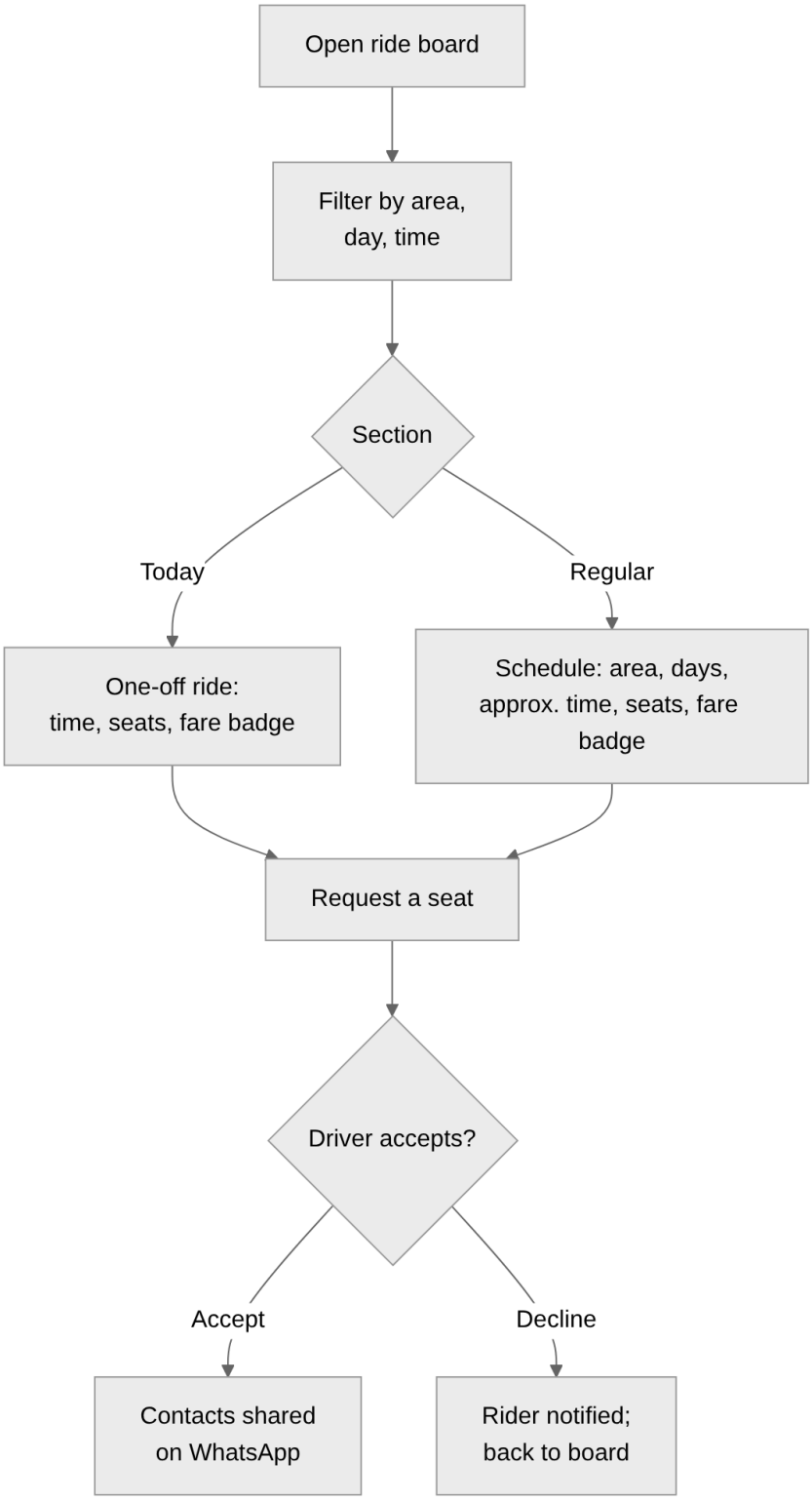
The status screen is where the rider spends most of her time, so it carries the most design weight (Scr 1–2).

## Flow 2: Driver responds to a ping



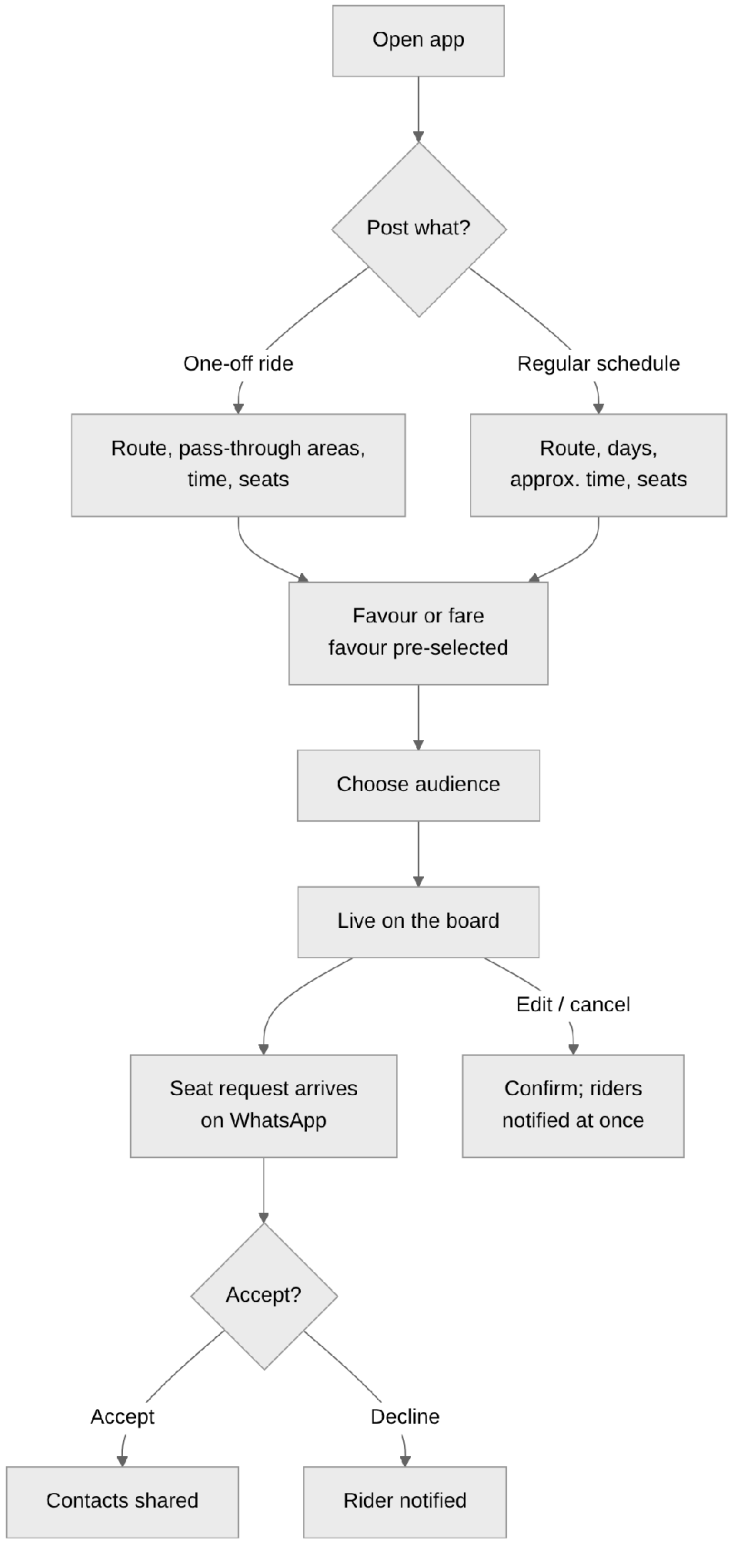
Goodwill drivers never need the app. The ping carries enough to decide on the spot (route, time, rider's first name, one trust signal); "See details" is optional. This is progressive disclosure: essentials first, more on demand.

**Flow 3: Rider uses the ride board**



"Today" holds one off offers, which expire after departure; "Regular" holds opt-in schedules. No phone number appears until the driver accepts (I7).

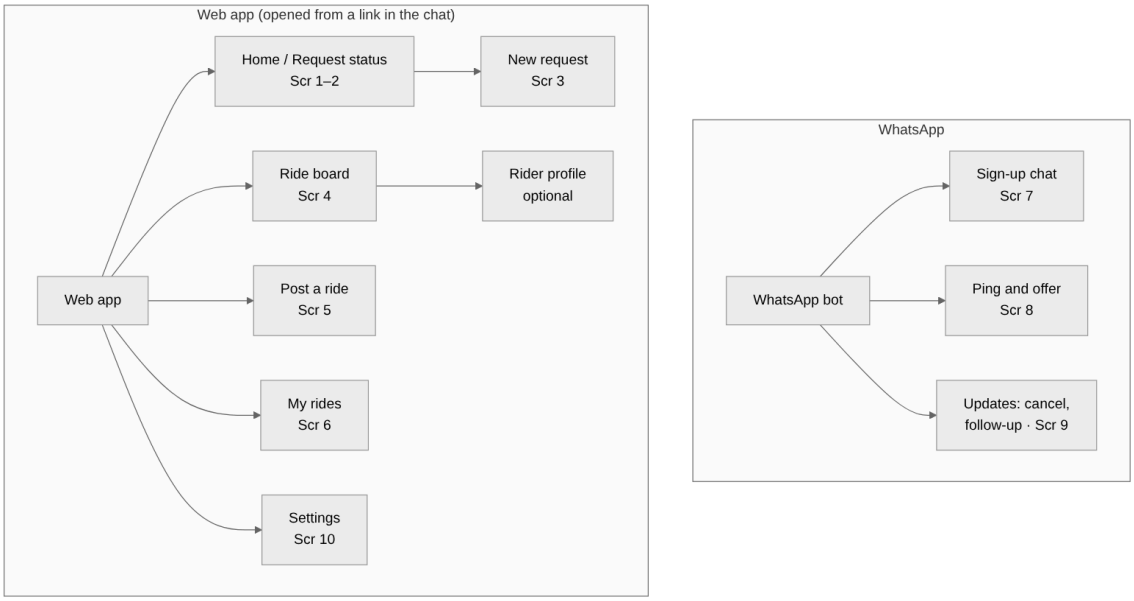
**Flow 4: Driver posts a ride from the app**



A one-off offer can also be posted with a quick WhatsApp message; the app adds schedules and audience settings.

## 6. Information architecture

The split follows commitment: WhatsApp for favours, the app for regular commutes. Goodwill drivers only use WhatsApp; riders and drivers with a schedule use the app.




*Site map. The rider profile opens from "See details" in a ping or a seat request.*

| Screen | Where | Who uses it | Purpose | Requirements | Wireframe |
|---|---|---|---|---|---|
| Sign-up (chat) | WhatsApp | Drivers, and riders who start there | Name, IBA email code, home area and pass-through areas | F7, F10, I4 | Scr 7 |
| Ping, offer, acceptance, handoff | WhatsApp | Drivers and riders | Answer in one tap, propose time, favour or fare, share contacts | F9, F12, I6, I7, I10 | Scr 8 |
| Updates | WhatsApp | Riders and drivers | Cancellations with a next step; after ride follow-up | F4, F6, F13 | Scr 9 |
| Quick commands | WhatsApp | Drivers | Post a one-off ride; pause pings | F11, I6 | — |

| Screen | Where | Who uses it | Purpose | Requirements | Wireframe |
|---|---|---|---|---|---|
| Home / Request status | App | Riders | Active request, offers to accept, no match options | F1, F6, F9, I1, I5, I9 | Scr 1–2 |
| New request | App | Riders | Route; "between" or "whenever" | F2, F8, I2, I4 | Scr 3 |
| Ride board and listing | App | Riders | "Today" and "Regular" rides, open requests, filters, fare badges; request a seat | F1, F2, F3, F11, I3, I7, I9 | Scr 4 |
| Post a ride | App | Drivers | One off ride or regular schedule; favour or fare; audience | F5, F11, F12, I2, I7, I8 | Scr 5 |
| My rides and seat requests | App | Drivers who post | See, edit or cancel posts; accept seat requests | F4, I5 | Scr 6 |
| Rider profile | App | Drivers (optional) | Opened from "See details" in a ping | I10 | — |
| Settings | App | Everyone who uses the app | Home and pass through areas, audience, notifications | F10, I6, I8 | Scr 10 |

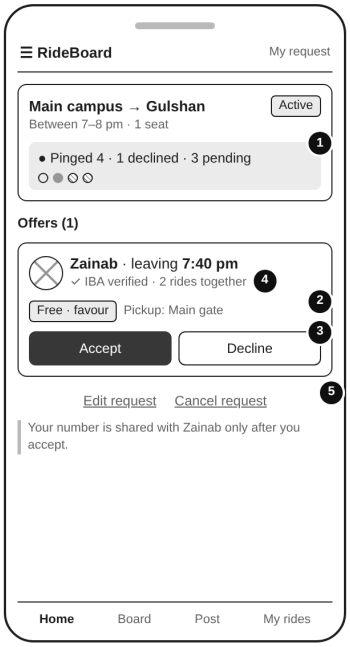
## 7. Wireframes

Low-fidelity, mobile first wireframes. They are grayscale on purpose, to focus feedback on structure and behaviour rather than visual style. Screens 1–6 and 10 are the web app; Screens 7–9 are WhatsApp mock chats, because a goodwill driver's whole experience happens there. The numbered markers on each screen link elements to requirements and HCI principles.

## Rider: request a ride

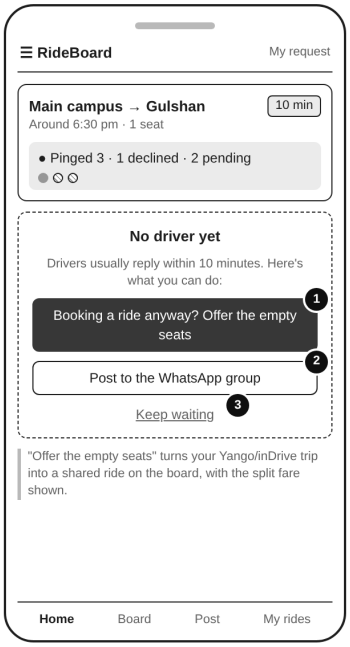
| Screen 1 · Request status (offer) | Screen 2 · Request status (no match) | Screen 3 · New request |
|---|---|---|

**Screen 1 · Request status (offer received)**
Home for a rider with an active request · F9, I1, I5, I9, I10



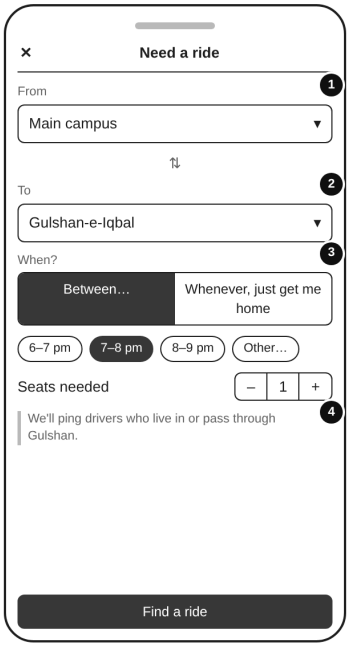

1. Live status line: who was pinged, declined, pending. Visibility of system status (I1)
2. Fare badge shown *before* accepting, never after. Error prevention (I9)
3. The ride is confirmed only when the rider accepts the proposed time. User control & freedom (F9)
4. One trust signal, with more on demand. Progressive disclosure (I10)
5. Edit or cancel without starting over. User control & freedom (I5)

**Screen 2 · Request status (no match yet)**
Shown after ~10 min with no offer · F6, I1




1. A dead end becomes a shared ride. Help users recover from errors (F6)
2. Fallback to the familiar channel. Match with the real world (F6, F7)
3. The rider decides when to give up, not the system. User control & freedom

**Screen 3 · New request**
A few taps, sensible defaults · F8, F10, I2, I4



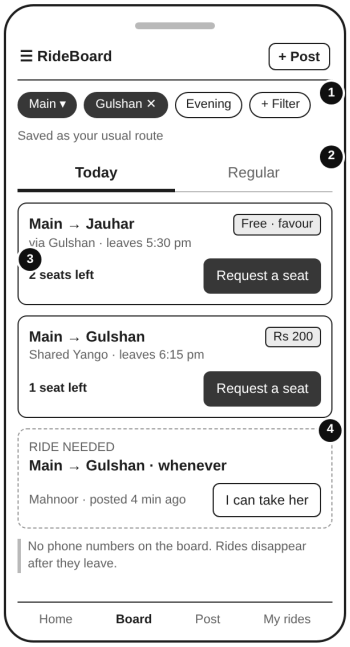

1. Campus pre-filled from last use. Efficiency of use (I2)
2. Area list uses the names students already say ("Gulshan", "Main"). Match with the real world (I4)
3. Time window or "whenever", because most plans are ad hoc. Flexibility (F8)
4. Says what will happen next. Visibility (F10)

## Board and driver posting

| Screen 4 · Ride board | Screen 5 · Post a ride | Screen 6 · My rides (cancel dialog) |
|---|---|---|

**Screen 4 · Ride board**

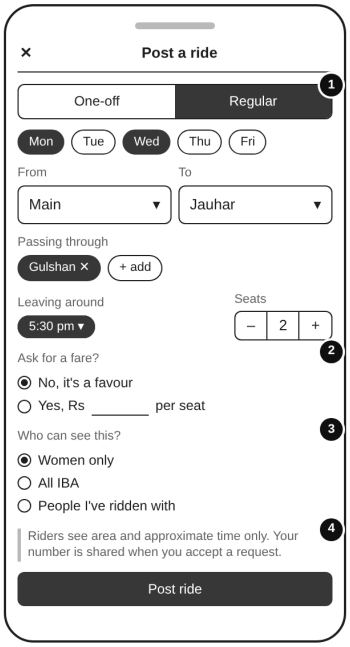
Browse offers and requests · F1, F2, F3, F11, I3, I7, I9




1. Filter chips update the list instantly; the last route is remembered. Recognition over recall (F2, I3)
2. "Today" one-off rides vs "Regular" schedules. Consistency (F11)
3. Seats left and fare on every card. Visibility of status (F3, I9)
4. Requests stay listed until filled or expired, instead of scrolling away. Visibility (F1)

**Screen 5 · Post a ride**

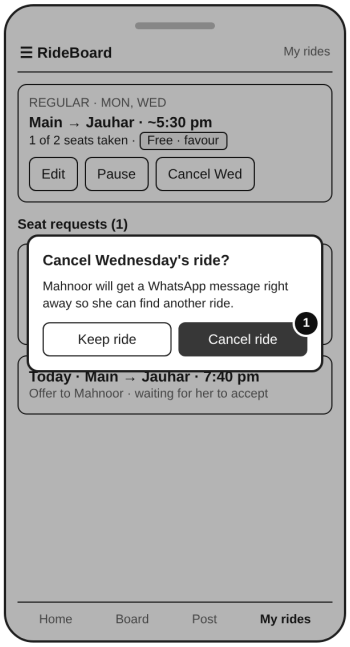
One-off or regular · F5, F11, F12, I2, I7, I8




1. Weekly schedule set once, not retyped every week. Efficiency of use (F5, F11)
2. Favour is pre-selected; a fare is optional. Defaults as a gentle nudge (F12)
3. Driver controls who sees the listing. User control (I8)
4. No street address or live location. Privacy by design (I7)

**Screen 6 · My rides (cancel dialog open)**

Driver manages posts and seat requests · F4, I5

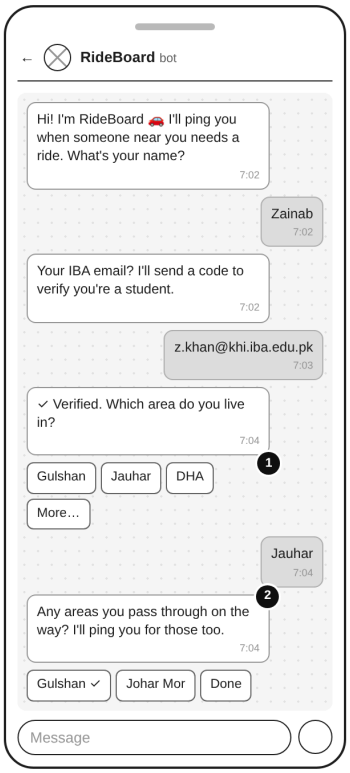



1. A confirmation step, and the rider is notified immediately. Error prevention + feedback (F4, I5)

## WhatsApp mock chats and settings

### Screen 7 · WhatsApp sign-up

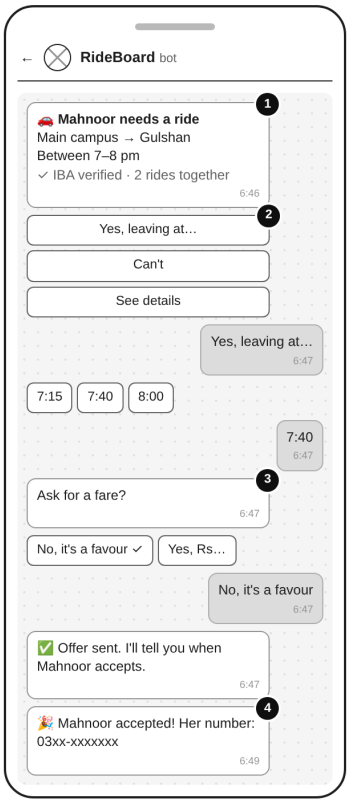
**Screen 7 · WhatsApp sign-up**
No app needed for goodwill drivers · F7, F10




1. Tap-to-answer area list, no typing. Recognition over recall (I4)
2. Home area + route areas, so drivers passing through also get pinged. Relevant notifications only (F10)

### Screen 8 · Ping and driver offer

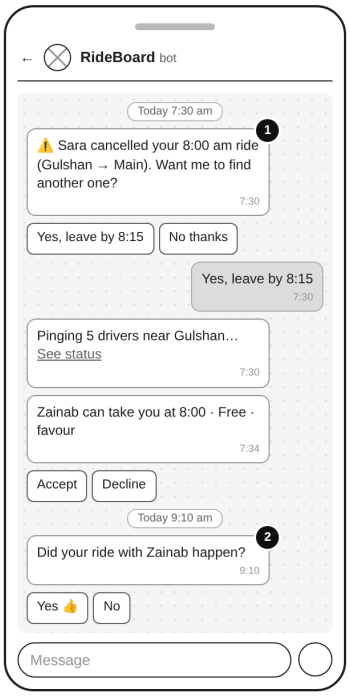
**Screen 8 · WhatsApp ping → driver offer**
Answer in one or two taps · F9, F12, I6, I7, I10




1. Enough to decide on the spot: route, time, name, one trust signal. Progressive disclosure (I10)
2. One-tap replies, no app needed. Efficiency of use (I6)
3. Fare question defaults to favour. Defaults as a nudge (F12)
4. Contacts are shared only after the rider accepts. Privacy by design (I7)

### Screen 9 · Rider updates

**Screen 9 · WhatsApp rider updates**
Cancellation and ride follow-up · F4, F6, F13

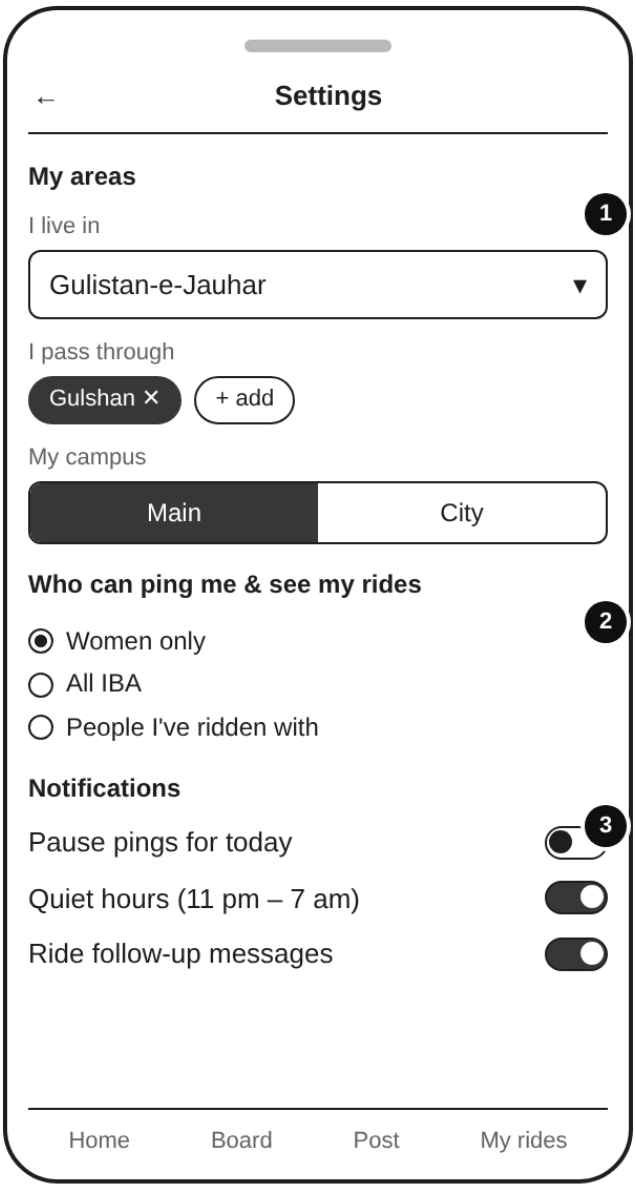



1. The cancellation is pushed immediately, with a next step attached. Feedback + error recovery (F4, F6)
2. One-tap follow-up closes the loop and keeps the board accurate. Feedback (F13)

Screen 10 · Settings

## Screen 10 · Settings

Areas, audience, notifications · F10, I8




1. Matching areas can be edited any time. User control (F10)
2. Audience setting, same wording as on Screen 5. Consistency (I8)
3. Drivers can stop interruptions. User control & freedom (I6)

## 8. HCI design principles

The table groups the requirements under the principles that justify them. Three decisions carry most of the design argument:

- **WhatsApp first, app second (F7, I6).** Members voted to stay on WhatsApp, and drivers are often busy when a request arrives (S2). So the high frequency, low effort actions (answering a ping, choosing favour or fare) happen in chat, and only lower frequency tasks (browsing, posting schedules, settings) need the app. This reduces switching costs and matches users' existing mental model.
- **The status screen as the centre of the rider experience (I1, F6).** P1's main failure was not knowing whether anyone had seen her request. Making system status visible, and turning the timeout into options rather than a dead end, targets the exact point where riders give up.
- **Favour by default, fare always visible (F12, I9).** The default mirrors current norms for ad-hoc rides; showing any fare before acceptance prevents the "trick" feeling of a late price.

| Principle | Source | Requirements | How it shapes the design |
|---|---|---|---|
| Visibility of system status | Nielsen (1994) #1; Norman (2013) | F1, F3, I1, I9 | The status line ("Pinged 4 · 1 declined · 3 pending") replaces the uncertainty that made P1 give up after ~10 minutes. The rider sees that the system is working instead of guessing. |
| Feedback | Norman (2013) | F4, F13, I1 | Every action gets a response in the channel the user is already in: offer sent, accepted, cancelled, and the after-ride check. |
| Match between system and the real world | Nielsen #2 | F7, I4 | Students already coordinate in WhatsApp and say "Gulshan" and "Main", so the bot lives there and the area list uses their words. |
| User control and freedom | Nielsen #3 | F6, F9, I5, I8 | Riders accept or decline proposed times and decide when to stop waiting; drivers choose their audience and can pause pings; posts can be edited or cancelled. |
| Consistency and standards | Nielsen #4; Shneiderman et al. (2016) | F11, I8 | The same ride card, fare badge and audience wording appear on every screen and in WhatsApp. |
| Error prevention | Nielsen #5 | I5, I9 | The fare is shown before accepting; cancelling a confirmed ride needs a confirmation step. |

| Principle | Source | Requirements | How it shapes the design |
|---|---|---|---|
| Recognition rather than recall | Nielsen #6 | F2, I2, I4, I10 | Chips, dropdowns and quick replies replace free text; the usual route is remembered. |
| Flexibility and efficiency of use | Nielsen #7 | F5, F8, I2, I3, I6 | One-tap WhatsApp replies for busy drivers, regular schedules for repeat commuters, and "whenever" for open-ended plans. |
| Aesthetic and minimalist design; cognitive load | Nielsen #8; Sweller (1988) | F10, I10 | Pings go only to relevant drivers and carry four facts (route, time, name, one trust signal); details are one tap away (progressive disclosure). |
| Help users recover from errors | Nielsen #9 | F6 | In the group chat, "no reply" is a dead end. Here it leads to three next steps, including sharing a ride-hail the rider books anyway. |
| Defaults as nudges | Thaler & Sunstein (2008) | F12 | "Favour" is pre-selected to match ad-hoc norms, while a fare remains one tap away for scheduled rides. |
| Privacy by design | Cavoukian (2009) | I7, I8 | Listings show area and approximate time only, and contacts are shared only on acceptance. |

## 9. Prototype

- **Platform:** mobile first web app, plus a WhatsApp bot that is simulated in the prototype.
- **Fidelity:** mid-fidelity: grayscale, real content and working navigation between screens.
- **Link**: https://github.com/h-e19/RideBoard-prototype

The prototype supports the following tasks for the Phase 3 task based evaluation:

| # | Task | Starts at | Success when | Tests | Measures |
|---|---|---|---|---|---|
| T1 | Request a ride home between 7 and 8 pm and accept the offer you receive | Scr 3 | Offer accepted | F8, F9, I1, I2 | Success, time, taps, errors |
| T2 | No driver has replied. Find out what you can do | Scr 2 | Chooses a fallback | F6, I1 | Success, time to notice |
| T3 | Find a regular Wednesday ride to Gulshan and request a seat | Scr 4 | Seat requested | F2, F11, I3, I9 | Success, taps, notices fare |

| # | Task | Starts at | Success when | Tests | Measures |
|---|---|---|---|---|---|
| T4 | As a driver, answer a ping: leave at 7:40 as a favour | Scr 8 | Offer sent | F9, F12, I6, I10 | Taps (target ≤ 3), time |
| T5 | Post a weekly Mon/Wed ride visible to women only | Scr 5 | Ride posted | F5, F11, F12, I7, I8 | Success, errors |
| T6 | Cancel a ride you offered | Scr 6 | Cancelled after confirmation | F4, I5 | Success, errors |

## 10. Iterative refinement

### Iteration 1 → 2: walkthrough with scenarios and Phase 1 evidence

The first version was checked by walking each scenario through it and against the Phase 1 findings. These changes came out of that review:

| # | Version 1 | Problem found | Change in version 2 | Evidence | Req. |
|---|---|---|---|---|---|
| 1 | A single departure time on requests | Rider plans are open-ended; most rides are ad hoc | Time window or "Whenever"; the driver proposes the time | Poll 14/17; S1; P2 | F8, F9 |
| 2 | Drivers had to open the app to reply | The driver is busy getting into her car | One-tap WhatsApp replies, including time and fare | S2; F7 vote | I6, F7 |
| 3 | Matching on home area only | A driver passing through Gulshan was never pinged | Home area plus pass through areas, at sign-up and in Settings | S3; P1 | F10 |
| 4 | Fare not handled | Paid and favour norms differ; a late fare feels like a trick | Favour by default; fare badge shown before accepting | Group norms | F12, I9 |
| 5 | Regular schedules showed phone numbers | Opening beyond a women only group exposes routines | Opt-in listings, area and approximate time only, contacts on acceptance; audience setting | P1, P3 | I7, I8 |
| 6 | No match state was a dead end | P1 and P4 fall back to the group or a solo ride | Three next steps: offer empty seats, post to group, keep waiting | P1, P4 | F6 |

| # | Version 1 | Problem found | Change in version 2 | Evidence | Req. |
|---|---|---|---|---|---|
| 7 | Ride outcome unknown | The board and status go stale after a match | One-tap after-ride follow-up | S1 walkthrough | F13 |

**Iteration 2 → 3: peer feedback on the wireframes**

| Participant | Screen | Feedback | Change made |
|---|---|---|---|
| R1 | Scr 3 : new request | Riders may need a flexible departure time instead of a fixed time window. | “Between…” and “Whenever, just get me home” appear side by side. Time chips provide quick choices for the “Between…” option. |
| R2 | Scr 8 : Whatsapp ping | Drivers need enough information to respond without opening the web app. | The ping shows the route, requested time window, rider’s name and trust signal. Drivers can select “Yes, leaving at…”, “Can’t” or “See details”, then choose a departure time and whether to request a fare. An “Offer sent” message confirms submission. |
| R3 | Scr 4: Ride board | Riders need to distinguish available rides from other students’ requests and see whether a ride is paid. | Available rides show “Request a seat”, seat availability and a fare badge such as “Free · favour” or “Rs 200”. Requests appear in a dashed box labelled “RIDE NEEDED”, with an “I can take her” button. |

## 11. Evidence gaps and limitations

- All Phase 1 data comes from the women only group, so the audience and trust requirements (I8, I10) rest on thin evidence for mixed users.
- There are no driver interviews yet. The driver persona and the supply side requirements (F10, F12, I6) are assumptions until tested in Phase 3.
- The ~10-minute abandonment threshold (F6, I1) comes from one participant (P1).
- "Women only" relies on gender self declared at sign-up; the prototype does not verify it.
- The sample is small: survey n = 26, 17 poll answers and four interviews (P1–P4).

- The WhatsApp bot is simulated. Real WhatsApp Business limits are not tested; for example, a message can carry at most three reply buttons, so longer choices such as the area list would use a list message.

## 12. References


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